Robinson Vidva

Computational Biologist
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SUMMARY

Computational biologist and bioinformatics scientist with more than 16 years of experience across systems biology modeling, multi-omics analysis, machine learning, and biostatistics for computational oncology, immuno-oncology, drug discovery, and pediatric neurodevelopment. Twelve years building and leading precision-oncology and immunology modeling platforms at Cellworks, three years running an independent consultancy, and, since 2025, bioinformatics and biostatistics for a neonatal neuroscience laboratory at Children's National Hospital. Integrates mutational, gene-expression, imaging, and clinical data to model treatment response and identify biomarkers. Managed a team of 12 scientists. First author of one and co-author of eight further peer-reviewed journal articles and 14 conference abstracts (150+ citations); named inventor on two patented method-of-use drug combinations (three grants, US and European) and three published patent applications; developer of open-source research tools, including a published lab-animal colony management platform.

TECHNICAL SKILLS

Computational biology and bioinformatics: NGS DNA and RNA analysis; bulk RNA-seq, genomics, and metabolomics; variant analysis (SNVs, CNVs); differential expression, pathway, gene-set enrichment, and cell-cell communication analysis; cancer genomics resources (TCGA, MSK-IMPACT, Beat AML); drug target and drug repurposing analysis

Systems biology and mechanistic modeling: mechanistic network models of signaling and gene regulation; ODE and stochastic pathway models; patient-specific drug-response models; MATLAB SimBiology and sensitivity analysis; drug repurposing and combination therapy design; model validation against in vitro, ex vivo, and in vivo data

Statistics and machine learning: generalized linear, mixed-effects, segmented, logistic, and modified Poisson models; survival analysis; stratification and sensitivity analyses; scikit-learn, TensorFlow, Keras, PyTorch

Programming and visualization: Python, R, SQL, MATLAB, C/C++, Unix shell, JavaScript/TypeScript, HTML/CSS, PHP; matplotlib, seaborn, ggplot2, Tableau; scientific presentations and dashboards

Clinical and multimodal data: quality control and data-issue review of de-identified clinical research datasets covering electronic health records (EHR), imaging (MRI, cranial ultrasound), laboratory, growth, and neurodevelopmental data; HIPAA and protected health information (PHI) handling; data-use agreements

Behavioral phenotyping: DeepLabCut, B-SOiD, BONSAI, OpenCV, MediaPipe; three-chamber sociability and Erasmus Ladder analysis; motor and gait phenotyping

Software engineering and infrastructure: full-stack web development; Git, GitHub, Jira; CI/CD; Snakemake; Docker and Singularity; database design; Linux server administration; LSF job scheduling; AWS; IoT integration for laboratory environments

EDUCATION

M.S., Data Analytics, Western Governors University, Salt Lake City, UT | 2025

M.B.A., Project Management, Sikkim Manipal University, Gangtok, India | 2020

B.Tech. (Bachelor of Technology), Biotechnology, Bharathidasan University, Tiruchirappalli, India | 2007

RESEARCH FOCUS

- Computational oncology and precision medicine: cancer genomics and NGS analysis; biomarkers of treatment response; patient-specific drug-response models; drug repurposing and combination therapy design; hematologic malignancies, especially acute myeloid leukemia (AML); cancer immunology.
- Immunology and inflammatory disease: systems biology models of macrophage, dendritic-cell, and T-cell signaling; mucosal and innate immunity; immune-modulating therapeutics; rheumatoid arthritis and inflammatory bowel disease.
- Pediatric and neonatal translational research: multi-domain clinical data (MRI, ultrasound, growth, neurodevelopmental assessments); preterm brain injury and cerebellar development; neonatal hypoxic-ischemic injury; necrotizing enterocolitis; sudden infant death syndrome; behavioral phenotyping and early rehabilitation.
- Open-source research software: a laboratory management platform (MyVivarium) and browser tools for enrichment analysis, drug-target exploration, drug interactions, and pose kinematics.

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Technician (Bioinformatics)

Children's National Hospital (Kratimenos Lab), Washington, DC | 03/2025 - Present

- Quality-check and integrate the de-identified clinical datasets (EHR, imaging, laboratory, metabolomic, and transcriptomic data) for two preterm-infant studies: cerebellar development after preterm birth (176 infants) and necrotizing enterocolitis and brain injury (181 infants).
- Run the statistical analyses relating clinical exposures to brain development in those cohorts, including more than 1,000 serial cranial ultrasound scans, using linear, mixed-effects, segmented, logistic, and modified Poisson models to evaluate cerebellar growth, MRI findings, and neurodevelopmental outcomes, with stratification and sensitivity analyses.
- Build and maintain reproducible RNA-seq pipelines for mouse and human cerebellar studies of prematurity insults, necrotizing enterocolitis, and sudden infant death syndrome; run differential expression, pathway, and Gene Ontology (GO) enrichment analyses.
- Quantify rodent behavior with DeepLabCut and B-SOiD in a mouse model of prematurity insults (maternal immune activation and neonatal hypoxia) and its early rehabilitation extension. Develop statistical and machine-learning models on Erasmus Ladder data on motor coordination from 160 preterm mice.
- Provide RNA-seq analysis for an astrocytoma immunotherapy study run by a collaborating laboratory in the center.
- Prepare figures, tables, and sequencing-data repository submissions for the laboratory's manuscripts, and check them against the deposited data.
- Co-authored a review on immune modulation in skin cancer, contributing the cancer-immunology content and figures (Frontiers in Immunology, 2025).

Career Break and Professional Development

Bayonne, NJ | 07/2022 - 02/2025

- Relocated from India to the United States on a dependent visa while awaiting work authorization; completed an accelerated M.S. in Data Analytics (Western Governors University, 2025) in a single term and the Imperial College London immunology course series (2023).
- Self-study in software engineering: full-stack web development, Linux server administration, and cloud deployment.

Founder and Lead Computational Biologist

Digirobi Solutions, Bengaluru, India | 09/2019 - 06/2022

- Ran an independent computational biology consultancy for commercial and academic clients in cancer therapeutics and research software.
- Assembled cancer genomics, drug, and nutrition databases from public and proprietary sources for a nutrigenomics platform and co-invented a cancer combination therapy later patented in the US and Europe.
- Developed a mechanistic model of Src kinase signaling in neonatal hypoxic neuronal injury (MATLAB SimBiology, sensitivity analysis), which resulted in a peer-reviewed paper (2022).
- Designed and built MyVivarium, an open-source colony-management platform with IoT sensing (published 2025).

Senior Lead Scientist

Cellworks Life, Bengaluru, India | 11/2017 - 09/2019

- Led a team of 12 scientists developing computational precision-medicine platforms for oncology.
- Using TCGA, MSK-IMPACT, and proprietary genomic, transcriptomic, and pharmacological data, designed and validated patient-specific drug-response models across solid and hematologic cancers. In the Beat AML study, the models' predicted sensitivity or resistance to a BET inhibitor agreed with the ex vivo IC50 classification for 93 of 100 AML patients (Leukemia Research, 2019). The same approach was applied to venetoclax, dasatinib, CDK4/6 and IDH inhibitors, and to chemotherapy response in pediatric AML (Blood, 2016 - 2018).
- Developed multi-cell computational models of epithelial, dendritic, and T cells whose predicted chemokine and cytokine responses matched laboratory co-culture measurements for 15 of 20 readouts after LPS stimulation and 16 of 20 after Pam3CSK4 stimulation (Scientific Reports, 2019).
- Assembled a cell-line resource (more than 500 lines with IC50 profiles) and more than 15 curated clinical datasets used to train, validate, and benchmark the models.

Lead Scientist

Cellworks Life, Bengaluru, India | 01/2014 - 10/2017

- Compiled a knowledge base of about 200 compounds covering targets, mechanisms, pharmacokinetics, and resistance, and developed a computational drug model for each compound within the disease platform.
- Co-invented the PTEN/TP53-targeted cancer combination therapy (US 10,098,880 B2), drafting the claims and handling prosecution through examiner objections.
- Delivered internal training on computational biology, immunology, and data analytics, evaluated therapeutic candidates with pharmaceutical partners, and mentored junior scientists.

Associate Lead Scientist

Cellworks Life, Bengaluru, India | 06/2008 - 12/2013

- Built and validated an eight-cell-type systems biology model of rheumatoid arthritis against in vivo data, identified two synergistic repurposed drug combinations, and showed that one reduced disease progression in a murine collagen-induced arthritis model.
- Contributed to peer-reviewed work on inflammatory skin and oral mucosal disease with collaborators, and to two patent applications and American College of Rheumatology presentations.
- Established the company's quality standards for model development and validation.

Associate Biomodeling Scientist

Cellworks Life, Bengaluru, India | 07/2007 - 05/2008

- Built mechanistic models of immune signaling in macrophages spanning more than 20 pathways and gene-regulatory networks and designed the perturbation-based validation approach adopted across the modeling platform.

PUBLICATIONS

Peer-reviewed

1. Sanidas G, Simonti G, Ghaemmaghami J, Woyshner K, Polyviou A, **Vidva R**, Lowe C, Vij A, Bittel D, Triantafyllou M, Wolff N, Cardenas R, Chen H, Almeida F, Sudhakar S, Mankad K, Koutroulis I, Sahlin Pettersen H, Stein-O'Brien G, Sidiropoulos DN, Gallo V, Kratimenos P. Intrinsic gestational timing governs human cerebellar development after preterm birth. Science. 2026;394(6823):549-55. doi:10.1126/science.aeb9095
2. Sanidas G, Simonti G, Ghaemmaghami J, Woyshner K, **Vidva R**, Bittel D, Wolff N, Triantafyllou M, Polyviou A, Byrd C, Lowe C, Salisbury H, Goldstein E, Sathyanesan A, Koutroulis I, Stein-O'Brien G, Sidiropoulos DN, Gallo V, Kratimenos P. Prematurity insults remodel cerebellar development and behaviour. Brain. 2026:awag363. doi:10.1093/brain/awag363
3. **Vidva R,** Raza MA, Prabhakaran J, Sheikh A, Sharp A, Ott H, Moore A, Fleisher C, Netherton H, Goldstein E, Pitychoutis PM, Nguyen TV, Sathyanesan A. MyVivarium: a cloud-based lab animal colony management application with real-time ambient sensing. Comput Struct Biotechnol J. 2025;27:612-23. doi:10.1016/j.csbj.2025.01.025
4. Kratimenos P, Vij A, **Vidva R**, Koutroulis I, Delivoria-Papadopoulos M, Gallo V, Sathyanesan A. Computational analysis of cortical neuronal excitotoxicity in a large animal model of neonatal brain injury. J Neurodev Disord. 2022;14(1):26. doi:10.1186/s11689-022-09431-3
5. Fischer CL, Bates AM, Lanzel EA, Guthmiller JM, Johnson GK, Singh NK, Kumar A, **Vidva R**, Abbasi T, Vali S, Xie XJ, Zeng E, Brogden KA. Computational models accurately predict multi-cell biomarker profiles in inflammation and cancer. Sci Rep. 2019;9(1):10877. doi:10.1038/s41598-019-47381-4
6. Drusbosky LM, **Vidva R**, Gera S, Lakshminarayana AV, Shyamasundar VP, Agrawal AK, Talawdekar A, Abbasi T, Vali S, Tognon CE, Kurtz SE, Tyner JW, McWeeney SK, Druker BJ, Cogle CR. Predicting response to BET inhibitors using computational modeling: a BEAT AML project study. Leuk Res. 2019;77:42-50. doi:10.1016/j.leukres.2018.11.010
7. Borgwardt DS, Martin AD, Van Hemert JR, Yang J, Fischer CL, Recker EN, Nair PR, **Vidva R**, Chandrashekaraiah S, Progulske-Fox A, Drake D, Cavanaugh JE, Vali S, Zhang Y, Brogden KA. Histatin 5 binds to Porphyromonas gingivalis hemagglutinin B (HagB) and alters HagB-induced chemokine responses. Sci Rep. 2014;4:3904. doi:10.1038/srep03904

8. Harvey LE, Kohlgraf KG, Mehalick LA, Raina M, Recker EN, Radhakrishnan S, Prasad SA, **Vidva R**, Progulske-Fox A, Cavanaugh JE, Vali S, Brogden KA. Defensin DEFB103 bidirectionally regulates chemokine and cytokine responses to a pro-inflammatory stimulus. Sci Rep. 2013;3:1232. doi:10.1038/srep01232

Review articles

1. Giannopoulos K, Karikis I, Byrd C, Sanidas G, Wolff N, Triantafyllou M, Simonti G, **Vidva R**, Koutroulis I, Theocharis S, Kratimenos P. Eph/ephrin-mediated immune modulation: a potential therapeutic target. Front Immunol. 2025;16:1539567. doi:10.3389/fimmu.2025.1539567

RESEARCH SOFTWARE

1. **MyVivarium, lead developer and maintainer** | 2021 - Present. Cloud platform for lab-animal colony management with QR cage cards and IoT environmental sensing (PHP, MySQL, JavaScript); in use at three laboratories (200+ animals); version 2 being deployed at the University of Dayton (2026). Source: github.com/myvivarium/MyVivarium. Published in Comput Struct Biotechnol J. 2025;27:612-23. doi:10.1016/j.csbj.2025.01.025.
2. **neokine, developer** | 2026. Client-side MediaPipe pose visualization in JavaScript with joint-angle, velocity, and asymmetry plots. Source: github.com/robinson-vidva/neokine. robinsonvidva.com/neokine.
3. **drugtargets, developer** | 2026. Drug-target exploration and repurposing tool in TypeScript, built on Open Targets and openFDA. Source: github.com/robinson-vidva/drugtargets. drugtargets.robinsonvidva.com.

Additional open-source tools: enrichlite, browser-based gene-set over-representation analysis (GO, Hallmark, Reactome); DrugInteract, client-side drug-interaction reference on openFDA and RxNorm. Details at robinsonvidva.com/tools.

PATENTS AND PUBLISHED PATENT APPLICATIONS

Granted

1. Vali S, Abbasi T, Stopka T, Minarik L, Singh NK, Usmani S, Radhakrishnan S, Sikora H, **Vidva R**, Pimkova K. Use of scientifically matched plant supplements combined with antineoplastic compounds for the treatment of hematological malignancies. US Patent 11,478,444 B2, granted 2022 Oct 25 (Brio Ventures LLC). European counterpart: EP 3,861,992 B1, Combination therapy comprising plant derived compounds and antineoplastic compounds for treatment of hematological malignancies, granted 2023 Sep 6 (Cannafamily AS).
2. Vali S, Usmani S, Sultana Z, Kumar A, Abbasi T, **Vidva R**. Combination of nelfinavir, metformin and rosuvastatin for treating cancer caused by aberrations in PTEN/TP53. US Patent 10,098,880 B2, granted 2018 Oct 16 (Cell Works Group Inc.).

Published applications

1. Vali S, Abbasi T, Fernandes P, Rajagopalan S, Alam A, **Vidva R**, Nair PR, Kapoor S, Radhakrishnan S, Sultana Z, Ramanujan KS, Tiwari KK, Kumar A, Singh NK, Agrawal AK, Talawdekar AA, Usmani S, Singh R. System and method for development of therapeutic solutions. US Patent Application 2015/0302167 A1, published 2015 Oct 22 (Cell Works Group Inc.).

2. Vali S, **Vidva R**, Nair PR, Fernandes P, Abbasi T, Radhakrishnan S. Compositions, process of preparation of said compositions and method of treating inflammatory diseases. US Patent Application 2015/0098993 A1, published 2015 Apr 9 (Cell Works Group Inc.).
3. Vali S, **Vidva R**, Nair PR, Fernandes P, Abbasi T, Radhakrishnan S. Compositions, process of preparation of said compositions and method of treating inflammatory diseases. US Patent Application 2014/0127295 A1, published 2014 May 8 (Cellworks Research India Pvt Ltd; Cell Works Group Inc.).

CONFERENCE ABSTRACTS

1. Sanidas G, Lowe C, Byrd C, **Vidva R**, Simonti G, Chandereng T, Gallo V, Kratimenos P. Necrotizing enterocolitis selectively disrupts pontine development in the preterm brain. Poster PSTR103.21. Society for Neuroscience Annual Meeting; 2025 Nov 16; San Diego, CA. [Abstract]
2. Simonti G, **Vidva R**, Sanidas G, Byrd C, Lowe C, Triantafyllou M, Wolff N, Chandereng T, Koutroulis I, Gallo V, Kratimenos P. Early postnatal rehabilitation mitigates motor, growth, and social deficits in perinatal brain injury. Poster PSTR019.10. Society for Neuroscience Annual Meeting; 2025 Nov 15; San Diego, CA. [Abstract]
3. Cogle CR, Abbasi T, Singh NK, Sauban M, Raman RK, **Vidva R**, Tyagi A, Talawdekar AA, Das A, Kornblau SM, Hoff FW, Horton TM, Vali S. AraC-daunorubicin-etoposide (ADE) response prediction in pediatric AML patients using a computational biology modeling (CBM) based precision medicine workflow. Blood. 2018;132(Suppl 1):4034. doi:10.1182/blood-2018-99-115775
4. Tyner JW, Druker BJ, Tognon CE, Kurtz SE, Drusbosky LM, Sahu D, **Vidva R**, Kapoor S, Azam H, Kumar R, Chickdipatti PG, Raveendaran NT, Gopi RP, Abbasi T, Vali S, Cogle CR. Predicting response to dasatinib using a computational model and its validation: a Beat AML project study. Blood. 2018;132(Suppl 1):1541. doi:10.1182/blood-2018-99-115678
5. Tyner JW, Druker BJ, Kurtz SE, Tognon CE, Drusbosky LM, **Vidva R**, Narvekar Y, Agrawal AK, Gera S, Prasad SA, Shyamasundar VP, Tunwer GS, Abbasi T, Vali S, Cogle CR. Predicting response to BET inhibitor in combination with palbociclib/sorafenib using a computational model: a Beat AML project study. Blood. 2018;132(Suppl 1):1540. doi:10.1182/blood-2018-99-115284
6. Drusbosky LM, Turcotte M, Castillo P, **Vidva R**, Gera S, Sauban M, Bhargav P, Deep P, Kapoor S, Abbasi T, Vali S, Tognon CE, Kurtz SE, Tyner JW, Druker BJ, Cogle CR. Predicting response to CDK4/6 inhibitors and combinations using a computational biology model: a Beat AML project study. Blood. 2017;130(Suppl 1):3909. [Abstract]
7. Drusbosky LM, **Vidva R**, Gera S, Lakshminarayana AV, Shyamasundar VP, Agrawal AK, Talawdekar A, Abbasi T, Vali S, Tognon CE, Kurtz SE, Tyner JW, Druker BJ, Cogle CR. Predicting response to BET inhibitors using a computational model and its validation: a Beat AML project study. Blood. 2017;130(Suppl 1):3908. [Abstract]
8. Drusbosky LM, **Vidva R**, Gera S, Das A, Tiwari KK, Joseph V, Mohapatra S, Lunkad N, Abbasi T, Vali S, Tognon CE, Kurtz SE, Tyner JW, Druker BJ, Cogle CR. Predicting response to IDH1/IDH2 inhibitors beyond IDH mutations using a computational model: a Beat AML project study. Blood. 2017;130(Suppl 1):3907. [Abstract]
9. Drusbosky L, Abbasi T, Vali S, Radhakrishnan S, Singh NK, Usmani S, Parashar D, **Vidva R**, Druker BJ, Tognon CE, Lo P, Tyner JW, Cogle CR. A genomic signature predicting venetoclax treatment response in AML identified by protein network mapping and validated by ex vivo drug sensitivity testing: a Beat AML project study. Blood. 2016;128(22):1713. doi:10.1182/blood.V128.22.1713.1713

10. Fischer C, Abbasi T, Vali S, Nair P, **Vidva R**, Tiwari K, Brogden KA. A predictive model of an oral inflammatory response. J Dent Res. 2014;93(Spec Iss A):1369. [Abstract]
11. Harvey L, Recker E, Raina M, Radhakrishnan S, Prasad S, **Vidva R**, Cavanaugh J, Vali S, Brogden KA. HBD3 both enhances and attenuates dendritic cell MAPK responses to HagB. J Dent Res. 2013;92(Spec Iss A):2820. [Abstract]
12. Singh G, **Vidva R**, Nair P, Radhakrishnan S, Fernandes P, Abbasi T, Refino C, Cruz JD, Vali S. Predictive software-based mathematical modeling: a novel approach to development of oral therapies for rheumatoid arthritis, validation in a murine collagen induced arthritis model. Arthritis Rheum. 2012;64(10 Suppl):S149-50. doi:10.1002/art.37735 [Abstract]
13. Vali S, Refino C, Cruz JD, **Vidva R**, Nair P, Radhakrishnan S, Fernandes P, Abbasi T, Singh G. Novel combination therapy of existing repurposed therapies designed by predictive software modeling shows profound impact on disease progression in a murine collagen induced arthritis model. Arthritis Rheum. 2012;64(10 Suppl):S148. doi:10.1002/art.37735 [Abstract]
14. Vali S, Hegedus C, Bhattacharjee C, **Vidva R**, Schmelzer K, Inceoglu B, Hammock BD. Insights into the anti-inflammatory action of the soluble epoxide hydrolase inhibitor through a systems biology based in silico modeling approach. FASEB J. 2008;22(1 Suppl):479.13. doi:10.1096/fasebj.22.1_supplement.479.13

TEACHING AND MENTORING

- Mentored more than 25 junior scientists at Cellworks Life and Digirobi Solutions, 2014 - 2022.
- Designed and delivered more than 15 internal training sessions on computational biology, data analytics, and modeling methods, 2014 - 2019.

PROFESSIONAL MEMBERSHIPS

International Society for Computational Biology (ISCB), 2024 - Present

ONLINE PROFILES

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Google Scholar: scholar.google.com/citations?user=xSxXsdAAAAAJ

GitHub: github.com/robinson-vidva | Web: robinsonvidva.com

PROFESSIONAL DEVELOPMENT

- Course series and specializations: immunology (Imperial College London, 2023; Rice University, 2020 and 2021); Cancer Biology (Johns Hopkins University, 2018); Design and Interpretation of Clinical Trials (Johns Hopkins University, 2019); Genomic and Precision Medicine (University of California San Francisco, 2017); Drug Development Product Management (University of California San Diego, 2017).
- DL-101 General Course on Intellectual Property and DL-204 Biotechnology and Intellectual Property, World Intellectual Property Organization (WIPO), 2012.

REFERENCES

Available on request.